

TED (15) – 4043 MODEL

Reg.No.

(REVISION -2015)

Signature

FOURTH SEMESTER DIPLOMA IN ENGINEERING/TECHNOLOGY

MICROCONTROLLERS AND INTERFACING

[Time : 3 hours]

(Maximum marks : 100)

PART – A

(Maximum marks : 10)

Marks

I Answer the following questions in one or two sentences. Each questions carries 2 marks.

1. State the function of ALE
2. MOV A,#25h is the example of which addressing mode of 8051
3. Write the Timer modes in 8051
4. What is the size of data memory in 8051?
5. What is the stepper motor? (5x2=10)

PART – B

(Maximum marks : 30)

II Answer any five of the following questions . Each questions carries 6 marks.

1. Explain the Stack of 8051?
2. Draw and explain the PSW Register of 8051?
3. Explain the Data Transfer instructions of 8051?
4. Explain the steps involved in interrupt processing of 8051?
5. Explain the PCON Register of 8051?
6. Explain the modes of operation of 8051 Timers?
7. Draw the interfacing Diagram LCD with 8051?

PART – C

(Maximum marks : 60)

(Answer one full questions from each unit. Each full question carries 15 marks)

UNIT - I

- III (a) Draw the Pin Diagram of 8051 (8)
(b) Compare Microcontroller and Microprocessor? (7)

OR

- IV (a) Draw the Block diagram of 8051 and explain? (8)
(b) Explain the Ports of 8051? (7)

UNIT – II

- V (a) Explain DIV and MUL Instructions with suitable examples. ? (8)
(b) Explain the IE register? (7)

OR

- VI (a) Write an ALP to find number of ones in a given byte? (8)
(b) Explain the Control Instructions of 8051? (7)

UNIT – III

- VII (a) Explain the Timer/Counters in 8051? (8)
(b) Explain the serial data transmission and reception in 8051 ? (7)

OR

- VIII (a) Explain different serial data transmission modes ? (8)
(b) Illustrate the function of TMOD and TCON special function registers (7)

UNIT –IV

- IX (a) Explain the interfacing of stepper motor with 8051? (12)
(b) Describe the keydebouncing ? (3)

OR

- X (a) Explain the Interfacing of water level indicator system with 8051? (15)

